

Anisakiasis, an underestimated infection: effect on intestinal permeability of Anisakis simplex-sensitized patients.



Anisakis simplex is a parasite that, if present in uncooked and contaminated saltwater fish, can invade the human gut. Two different clinical situations are recognized: the first, known as a gastrointestinal disease, varying from an asymptomatic episode to vomiting and diarrhea, and the second, classified as an adverse reaction to food, characterized by a wide spectrum of allergic reactions like rhinitis, conjunctivitis, or even anaphylaxis causing hypotension and/or shock. The intestinal epithelium, the major defense system against external molecules, represents an open gate for toxins and allergens if its protective function is compromised. Previous data have demonstrated a strict relationship between an altered intestinal permeability (I.P.) and worsening of the clinical manifestations in patients with adverse reactions to the food. In this article we evaluated the sensitization to A. simplex among patients who referred clinical symptoms of allergy. All subjects underwent commonly used alimentary skin prick test for food allergens, to which Ani s1, an A. simplex allergen, was added. In addition, in A. simplex-sensitized subjects, I.P. was determined upon their enrolment to the study (time 0) and after 6 months of consuming a raw fish-free diet (time 6). Five hundred and forty subjects were screened, and 170 had a positive skin prick test, 87 (51.2%) of whom were positive to Ani s1. Increased I.P. was evidenced in A. simplex-sensitized subjects with worse clinical symptoms, which receded after 6 months' elimination of raw seafood. With our data we demonstrated that the alimentary habit to eat raw fish represents a high risk for the integrity of the intestinal mucosa, and we suggest that this pathological situation may constitute an ideal, under-estimated, open gate for molecules that predispose to other, more important pathologies.